The Primordial Quark Theory: A Derivative Approach to Matter

(The Origin of Subatomic Particles)

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June 11, 2026

Abstract

This paper proposes a mathematical and conceptual framework for the hypothesis that quarks represent the unified "raw matter" ($\mathcal{Q}_{raw}$) prior to the Big Bang. The theory posits that electrons, despite possessing an independent negative charge, are fundamentally "physical derivatives" resulting from the dissociation of this primordial raw matter during cosmic expansion.

1. The Big Bang Dissociation Equation

Assuming the universe's starting point was not merely energy, but a unified "raw quark mass," the dissociation caused by the Big Bang (E_{BB}) can be expressed as:

$$\mathcal{Q}_{raw} \xrightarrow{E_{BB}} \sum_{i=1}^{n} q_i + \sum_{j=1}^{m} e_j^- \tag{1}$$

Where:

- $\mathcal{Q}_{raw}$: The quark as a raw, undivided primordial mass.
- E_{BB}: The Big Bang energy (the catalyst for dissociation).
- q_i: The resulting fractional-charged quarks after expansion.
- e_j^-: The electrons emerged as "derivatives" with an independent charge of negative one.

2. Charge Relationship and Independence

Although the electron maintains an independent charge ($q_e = -1$), this theory classifies it as an elementary particle derived from the primordial raw material. This relationship can be modeled as an energetic derivative function:

$$e^- = \frac{\partial(\mathcal{Q}_{raw})}{\partial(E_{BB})} \tag{2}$$

This mathematical representation illustrates that the electron (e^-) is not contained within the quark, but is the result of the transition (derivation) of the raw matter relative to the cosmic expansion energy.

This formulation serves as a theoretical framework for further discussion and development in particle cosmology.